

# 0072 — HEAD's state is said on the wire, because an absent branch name means two opposite things

**Status:** Accepted — implemented, tested, and driven in the live app **Date:** 2026-08-24 **Issue:** [#473](#) · the gap [ADR 0071](#) recorded and deferred

## Context

ADR 0071 settled that a HEAD which resolves to nothing is **recorded** (`HeadAtEvent::Unresolvable`) rather than **badged**. It also recorded, plainly, that this was *option B for everything the user sees* — because `HeadAtEvent` never leaves the journal-capture path, and the live payload could not carry the distinction.

That left one field doing two jobs:

| Repository state                          | <code>head_branch</code> | What the topbar drew |
|-------------------------------------------|--------------------------|----------------------|
| detached at a real commit — <b>normal</b> | <code>null</code>        | nothing              |
| HEAD resolves to nothing — <b>broken</b>  | <code>null</code>        | nothing              |

The user most needs to be told in the second row, and the app was at its quietest there. A dangling HEAD is what a repository looks like mid-recovery, after a bad manual ref write, or when the object a detached HEAD pointed at has been garbage-collected.

# Decision

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## D1 — The state is a field, not an inference

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`HistoryFrame` gains `head_state: HeadState` — `OnBranch`, `Detached`, `Unborn`, `Unresolvable`, `Unknown`.

After ADR 0071 the client *could* infer it: no branch name **and** no `Head` ref means unresolvable, because a dangling HEAD is no longer badged. That inference is rejected. It is the shape ADR 0068 refuses — a state read from the absence of two other things is a state nobody can find when it is wrong, and it silently couples the topbar to a badging rule that exists for unrelated reasons.

## D2 — The vocabulary lives in the protocol crate

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`git-vista-protocol` is pure and wasm-safe and deliberately carries no dependency on `git-vista-core`; the envelopes are generic over their domain types for exactly that reason. So `HeadAtEvent` — which lives in core, and is journal-shaped, carrying oids — is not the type to put on this wire.

`HeadState` is its own small enum with no oids: the frame already carries `head_branch` and the refs. The server maps between them in four lines, with the same four-way reading `read_refs_at` applies (ADR 0071).

## D3 — `Unreadable` is deliberately absent

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When `.git/HEAD` itself will not read, `read_history_materials` fails the whole request: the user gets an error, not a frame. There is no frame in which that state could be reported, so there is no variant for it. **A variant that cannot occur is vocabulary nobody can trust** — the same reasoning ADR 0070 applied to capture fields that must not claim more than they observed.

## D4 — A missing field reads as `Unknown`, never as healthy

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`#[serde(default)]` with `Unknown` as `Default`. The frontend ships compiled into the server, so skew is narrow — but a browser holding a stale bundle is a real state on this box, and the failure must be "say nothing", never "claim HEAD is fine".

## D5 — Only the broken state earns a notice

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`Detached` is ordinary and deliberate and gets no label. A warning that fires on healthy repositories is a warning nobody reads, and labelling every branchless HEAD would have satisfied a naive test while making the app worse.

**Reviewed in use and kept, 2026-08-25.** After driving the shipped build against a deliberately-broken repository, the owner was offered two alternatives to the silence — a neutral `detached` label, and a `detached at <short oid>` form that says where you actually are — and chose to keep it silent. The silence is therefore a **ratified decision, not an unfilled gap**. A later session finding a branchless HEAD that renders nothing should not read it as an oversight and helpfully close it.

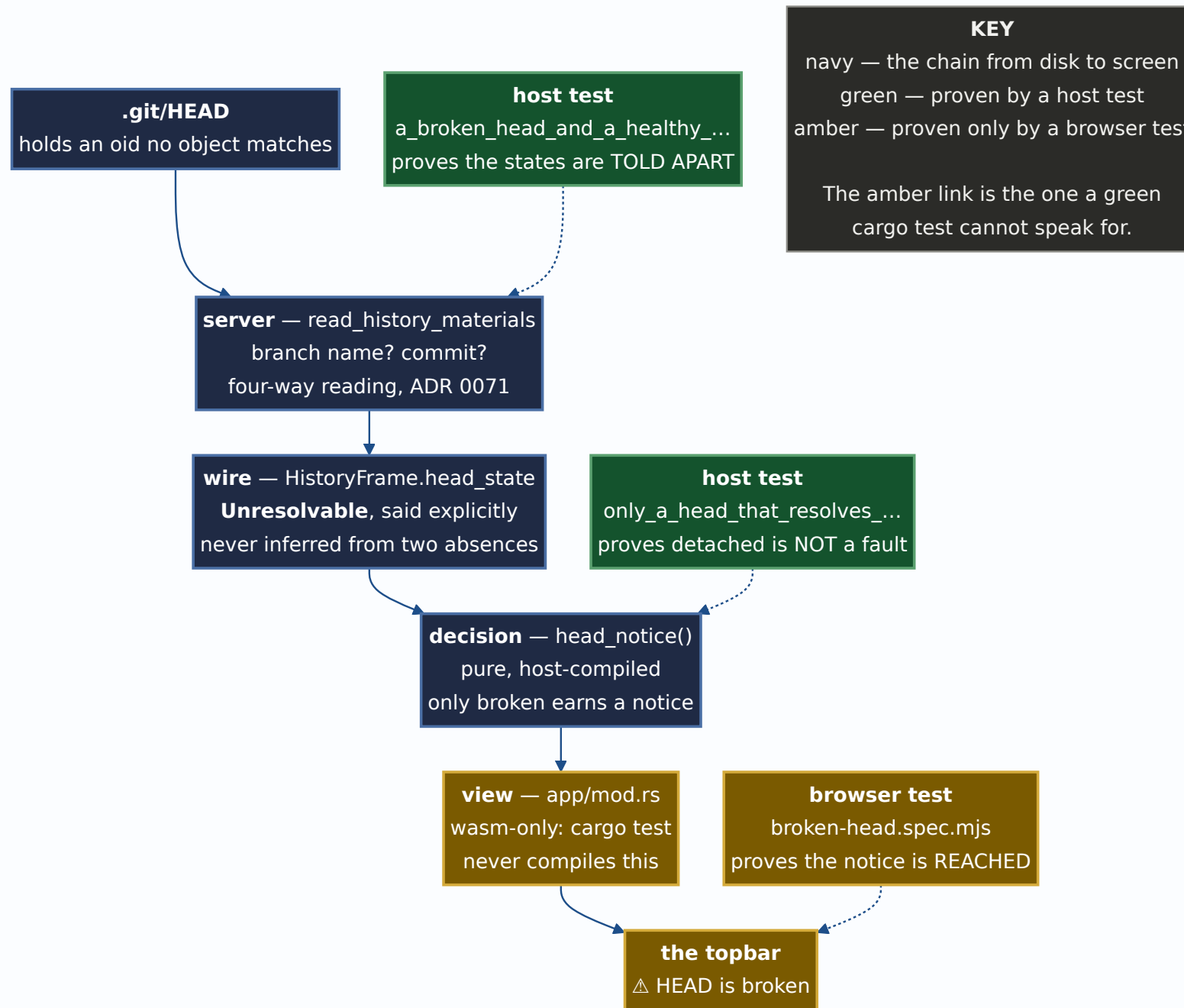
## D6 — The decision is host-tested; the wiring is browser-tested

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`head_notice` is a pure function in its own module, **not** a `match` inside the view: `mod app is #[cfg(target_arch = "wasm32")]`, so anything decided inside it can never be host-tested — and "renders nothing" is precisely the defect being fixed. The same posture as `hook_policy_disclosure`.

A host test cannot prove the decision is *reached*, so a fifth browser fixture — a repository whose `.git/HEAD` holds an oid nothing resolves — proves the notice appears there and **does not** appear on a healthy repository.

The diagram at the end of this section shows where each layer is proven.



# Alternatives considered

| Option                                                | Why not                                                                                                                                                                                |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Infer it client-side from "no branch and no HEAD ref" | Works today only because ADR 0071 stopped badging a dangling HEAD. Couples the topbar to a badging rule that exists for other reasons, and ADR 0068 refuses states read from absences. |
| Put HeadAtEvent on the wire                           | It lives in git-vista-core, which the protocol crate deliberately does not depend on, and it is journal-shaped — it carries oids the frame already has.                                |
| A head_unresolvable: bool                             | Answers one question and forecloses the others. The issue's own requirement was that the three null rows be distinguishable, not that one be flagged.                                  |
| Label detached too                                    | Cries wolf on an ordinary state. See D5.                                                                                                                                               |
| Decide inside the view                                | Untestable on the host, which is how the original defect survived. See D6.                                                                                                             |

# Consequences

## Good

- A broken repository says it is broken, in the place the user is already looking.

- The three states that share `head_branch: null` are distinguishable on the wire, so future surfaces need no new server work.
- The wire change is additive: one field, defaulted. Old clients ignore it; new clients tolerate its absence.

## Bad, and accepted

- One more field on a frame fetched once per view. It is a short enum string.
- `Unreadable` remains unreportable in a frame (D3). If that request-level error is ever softened, this enum will need the variant — and that is the moment to add it, not before.
- The notice is text in the topbar, not a guided explanation. #141's status-chip panel is where a fuller "here is what happened and what to do" belongs.
- **The feature spans two build artifacts, so verifying one proves nothing.** The state is decided by the server and drawn by the wasm bundle. On the night this shipped, the bundle was rebuilt and confirmed to carry the notice text while `target/release/git-vista-server` — the binary the systemd unit actually runs — was two days old and contained no `head_state` at all. Every frame then decodes as `Unknown`, which by D4 correctly draws nothing, so the app is silent in exactly the way it is silent when everything is fine. A green gate says nothing about which binary is running: check both halves, or check the live API.

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## Evidence

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- Golden wire fixture regenerated deliberately; the diff is exactly one additive line, `"head_state": "on_branch"`.
- Two mutations, both `caught`, conclusive, failing in **different layers**:

| Mutation                                        | Fails on                                      |
|-------------------------------------------------|-----------------------------------------------|
| server maps (None, None) to Detached            | the states collapse — assert_ne! on the frame |
| head_notice returns the notice for Detached too | an ordinary state reported as a fault         |

- Browser: the notice appears on the broken fixture and is absent on the healthy one. The second assertion is the load-bearing one.
- **Driven by a human, both cases, 2026-08-25 01:35.** A demo repository (four commits, a side branch, main intact, .git/HEAD holding a well-formed oid with no object behind it) showed the red notice while the graph below it still rendered every commit — the readable half surviving, which is the state the notice has to be legible against. Returning to a healthy repository, the notice was **absent**. That second observation is the one no amount of green CI settles.
- Live API, same run: POST /api/select onto the broken repository, then GET /api/frame, returning head\_state = "unresolvable" with head\_branch = null — the two facts D1 exists to separate, observed together on the wire.
- ./dev gate green, all six legs.